Python Code

```python
"""#question 1
nums = [2,7,6,3,7]
target = 9

i = 0
j = 1

while i < len(nums) - 1:
    if j == len(nums):
        i += 1
        j = i + 1
    elif nums[i] + nums[j] == target:
        print(i, j)
        j += 1
    else:
        j += 1"""
"""#question 2
l1 = ["1","2","4"]
l2 = ["1","3","4"]
i=0
x=""
while i<len(l1):
    x+=l1[i]
    i+=1
y=""
j=0
while j<len(l2):
    y+=l2[j]
    j+=1
z=int(x)+int(y)
print(z)"""
"""#question 3
s = "Leetcode"
max_str=""
j=0
while j<len(s):
   i=j
   x=s[j]
   while i<len(s)-1:
      if s[i+1] not in x:
       x += s[i+1]
      else:
        break
      i+=1
   if len(x)>len(max_str):
     max_str=x #this changes every time x becomes grreater than max_str because max_str is storing the lagest value each time very cool couldnt think myself but got the logic
   j+=1
print(max_str)
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print(len(max_str))"""
"""#question 4
numbers_1 = [0,1,0,3,12]
numbers_2 = [0,4,1,100,200]#python automatically sorts till like 20 after that it becomes unsorted as it should be because of sets

#final_list = list(set(numbers_1 + numbers_2))#set removes duplicates another good idea i didnt have sadly  THEY DIDNT ASK TO REMOVE DUPLICATES

final_list = numbers_1 + numbers_2
final_list.sort()
print(final_list)
x=len(final_list)%2
if x==0:      #it will be better to do x%2==1 as the case for odd is faster to check first but its okay
    median=(final_list[len(final_list)//2]+final_list[(len(final_list)//2)-1])/2
    print(median)
else:
    median=final_list[len(final_list)//2]
    print(median)"""
"""#question 5   same as question 3 but with palindrome check and i still couldnt do it myself understood the logic very quickly tho
#this solution was wy too slow
word = "abchfwijughuiwji3iiioddbbddoiewklfniwnailfhuoiwdcbe"
max_str = ""
j = 0

while j < len(word):
    i = j
    x = word[j]

    while i < len(word) - 1:
        x += word[i + 1]
        if x == x[::-1] and len(x) > len(max_str):
            max_str = x
        i += 1

    j += 1

print("Longest Palindrome:", max_str)

#optimal way to do it

start = 0
end = 0

i = 0
while i < len(word):

            # odd-length palindrome
            left = i
            right = i
            while left >= 0 and right < len(word) and word[left] == word[right]:
                if right - left > end - start:
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                    start = left
                    end = right
                left -= 1
                right += 1

            # even-length palindrome
            left = i
            right = i + 1
            while left >= 0 and right < len(word) and word[left] == word[right]:
                if right - left > end - start:
                    start = left
                    end = right
                left -= 1
                right += 1

            i += 1

print(word[start:end + 1])"""
"""#question 6  Good one i couldnt think of myself ahahahahahahahahahahaah i did think of a way using matix were you need to make many square matices but this is way better and i dont know how to write the code for that way  i did think of a few other ways didnt worrk tho one ws of adding spces which is similar to this corect way but i just couldnt make it work
              # the mtrix method i stated was something like this you solve this problem by making a square matrix many times so like the first colum will have the complete elments then if i=j meaning the digonal elements of the mtrix only then taking the word and rest spacing empty lso can start from i=n-1 n is the number of rows aand decreasing so on so you will have like a square matrix having first and last column filled and rest diagonal elements giving you a zig zag then because the wod couldnt be completed in just one square matrix then you make a new one whose first column will be the last column of the matrix just before it nd repeting it till the word is complete aand to print the answet you tke all the elements of the ow of first matrix added with the aall the elements exxcept the first column elements of every net meti doing the same fo ll the other rows to get the answer
word = "PAYPALISHIRING"
num = int(input("enter number of rows: "))
if num == 1:
    print(word)
result = ""
skips = 2 * num - 2
for r in range(num):
    for i in range(r, len(word), skips):
        result += word[i]
        if r != 0 and r != num - 1:
            j = i + skips - 2 * r
            if j < len(word):
                result += word[j]


print(result)"""
"""#question 7
nums = -1283890490
if nums<0:
    rev = -int(str(-nums)[::-1])
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else:
    rev = int(str(nums)[::-1])
if rev < -2**31 or rev > 2**31 - 1:
    rev = 0
print(rev)

    #alternative way using less memory

s = str(nums)
if s[0] == '-':
    reversed = '-' + s[:0:-1]
else:
    reversed = s[::-1]

num = int(reversed)

if -2**31 <= num <= 2**31 - 1:
    print(num)
else:
    print(0)"""
"""#question 8     good one did do it almost myself but my code had errors and i couldnt debug it properly
s = "    +462ab c"
s = s.lstrip()  # Remove leading whitespace
integer_str = ''

if len(s) == 0:
    integer_str = '0'

for i in range(len(s)):
    if i == 0 and s[0] == '+':
        continue
    elif i == 0 and s[0] == '-':
        integer_str += '-'
    elif s[i].isdigit():
        integer_str += s[i]
    else:
        break
if integer_str == '' or integer_str == '-':
    integer_str = '0'
print(integer_str)"""
"""#question 9
num = -121
str_num = str(num)
print(str_num == str_num[::-1])"""
"""#question 10     my god this question was hard couldnt even move it had to watch a video and still coulnt do it then had to ask chatgpt to do it for me and explain every single line
s = "ccaabpde"
p = "c*a*b..."
m = len(s)
n = len(p)

dp = [[False] * (n + 1) for _ in range(m + 1)]
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dp[0][0] = True

for j in range(2, n + 1):
    if p[j - 1] == '*':
        dp[0][j] = dp[0][j - 2]

for i in range(1, m + 1):
    for j in range(1, n + 1):

        if p[j - 1] == s[i - 1] or p[j - 1] == '.':
            dp[i][j] = dp[i - 1][j - 1]

        elif p[j - 1] == '*':
            dp[i][j] = dp[i][j - 2]

            if p[j - 2] == s[i - 1] or p[j - 2] == '.':
                dp[i][j] = dp[i][j] or dp[i - 1][j]

result = dp[m][n]
print(result)"""
"""#random question i found on insta by codingblocks day4/200 reel
nums=[2,5,6,7,4,9,3,1,8]
for i in range(len(nums)):
    if i%2==1:
        if nums[i]<nums[i-1]:
            nums[i],nums[i-1]=nums[i-1],nums[i]
        if i+1<len(nums) and nums[i]<nums[i+1]:
            nums[i],nums[i+1]=nums[i+1],nums[i]
print(nums)"""
"""#random question i found on insta
array=[1,2,1,3,5,7,7,4,9]
array.sort()
k=0
for i in range(len(array)):
    if array[k]!=array[i]:
        k+=1
        array[k]=array[i]
print(array[:k+1])"""
"""#random question
arr= ["bana","banaya","apple","grape","banrra","banana","bandana","banann"]
target="banana"
for i in range(len(arr)):
    count=0
    if len(target)==len(arr[i]):
        for j in range(len(target)):
            if target[j]!=arr[i][j]:
                count+=1
                if count>1:
                   count=0
if count==1:
    print(arr[i])"""
"""#question 11
heights = [1,3,2,4,5,3,2,1]
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max_value = 0
for i in range(len(heights)-1):
    for j in range(i+1,len(heights)):
        area = min(heights[i], heights[j]) * (j - i)
        max_value = max(max_value, area)
print(max_value)
#two pointer approach
left = 0
right = len(heights) - 1
max_area = 0

while left < right:
    area = min(heights[left], heights[right]) * (right - left)
    max_area = max(max_area, area)
    if heights[left] < heights[right]:
        left += 1
    else:
        right -= 1

print(max_area)"""
#question 12 int to roman numeral
number= str(input("enter a number: "))
for i in range(len(number),0,-1):
    if i>=4:
        print("M"*(int(number[len(number)-i])))
    elif i==3:
        if int(number[len(number)-i])==9:
            print("CM",end="")
        elif int(number[len(number)-i])>=5:
            print("D"+"C"*(int(number[len(number)-i])-5),end="")
        elif int(number[len(number)-i])==4:
            print("CD",end="")
        else:
            print("C"*(int(number[len(number)-i])),end="")
    elif i==2:
        if int(number[len(number)-i])==9:
            print("XC",end="")
        elif int(number[len(number)-i])>=5:
            print("L"+"X"*(int(number[len(number)-i])-5),end="")
        elif int(number[len(number)-i])==4:
            print("XL",end="")
        else:
            print("X"*(int(number[len(number)-i])),end="")
    elif i==1:
        if int(number[len(number)-i])==9:
            print("IX",end="")
        elif int(number[len(number)-i])>=5:
            print("V"+"I"*(int(number[len(number)-i])-5),end="")
        elif int(number[len(number)-i])==4:
            print("IV",end="")
        else:
            print("I"*(int(number[len(number)-i])),end="")
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